Abdelhakim Amer

Industrial PhD student at Aarhus University/EIVA
MSc. in Mechanical Engineering

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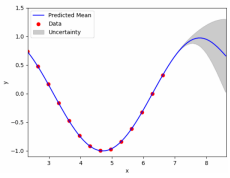
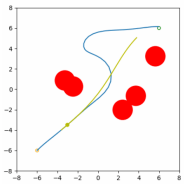
Profile

I am passionate about robotics, autonomous systems, learning-based control, and field robotics for industrial applications. My research focuses on leveraging **machine learning** and **optimization** to improve robot trajectory tracking and enhance control, planning, and dynamics modeling in autonomous mobile robots. I'm driven by the challenge of advancing autonomous systems and bridging **theory** with practical solutions for **real-world applications**.



Gaussian Processes



Model Predictive Control

Advanced Simulation

Academics

I hold a bachelor's in Mechanical Engineering with a focus on mechatronics and design, and a master's in High Tech Mechatronics Systems from Delft University of Technology, specializing in numerical methods, modeling, and control. My interest in autonomous systems grew through a course on micro aerial vehicle navigation, leading me to the AIR Lab at Aarhus University, where I contributed to a wind turbine inspection project by developing control algorithms for aerial vehicles and gaining expertise in ROS and C++. I am currently pursuing a PhD in learning-based optimal control at EIVA, with a focus on model uncertainties.

2022 - now	EIVA A/S, Denmark	**PhD**, Robotics and A.I
2024 (1 month)	DFKI, Germany	**Visiting Scholar**, Underwater Robotics Research Group
2018 - 2021	Delft University of Technology, The Netherlands	**Master**, Mechanical Engineering
2013 - 2018	The American University in Cairo, Egypt	**Bachelor**, Mechanical Engineering

Work Experience

Prior to my PhD, I completed a 3-month internship at Royal IHC, where I worked on developing algorithms for path planning and control. After graduation, I spent a year at the AIR Lab at Aarhus University, working on an autonomous wind turbine inspection project, where I gained experience in software and hardware development for drones. I then began my industrial PhD at EIVA, where I am finalizing my research in learning-based optimal control, where I have developed state-of-the-art methods for planning and control of autonomous underwater vehicles, focusing on addressing model uncertainties.

EIVA	2022 - 2025	EIVA A/S	PhD Researcher in Robotics and AI
	2021 - 2025	Aarhus University	Graduate Teaching Assistant in Control and Robotics
AIRLab	2021 - 2022	AIR Lab, Aarhus University	Research Assistant in Autonomous Systems
IHC	2020	Royal IHC	Dynamics and Control Graduate Intern (3 months)

Programming Experience

In recent years, I have primarily developed in Python and C++ for robotics applications, particularly in model-based and learning-based control, planning, and simulation. I have extensive experience with the ROS framework for integrating robotic systems. Additionally, I have worked with Unreal Engine for developing high-fidelity robotic simulations.

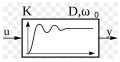
Python	C++	ROS	Unreal Engine	MATLAB
★★★★★	★★★★☆	★★★★★	★★★☆☆	★★★★☆

Teaching and Supervision

As a Teaching Assistant at Aarhus University, I contributed to the Control Theory and Autonomous Mobile Robotics courses, providing support to students and assisting with course material.

Teaching Assistant Roles

Autonomous Mobile Robots

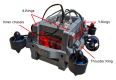


Control Theory

Additionally, I have supervised several Bachelor's and Master's theses, guiding students through projects in areas such as ROTV design, underwater vehicle modeling, and drone design.

Supervision of Thesis/Projects

Fall 2024 MSc Thesis: Underwater Vehicle Modeling and Control using PINNs


Spring 2023 R&D: 3-D printing ship propeller cleaning ROV design

Spring Fall BSc Thesis: Seabed Visualizing ROTV

Fall 2022 BSc Project: Small Inspection Drone Design and Build

Publications

1. **Amer, A.**, Falsegar, D., Brodskiy, Y., & Sarabakha, A. (2025). *Modelling of Underwater Vehicles using Physics-Informed Neural Networks with Control*. Submitted to the International Joint Conference on Neural Networks (IJCNN), under review.

2. **Amer, A.**, Mehndiratta, M., Brodskiy, Y., & Kayacan, E. (2025). *Empowering Autonomous Underwater Vehicles Using Learning-based Model Predictive Control With Dynamic Forgetting Gaussian Processes*. IEEE Transactions on Control Systems Technology.

3. Liang, W., **Amer, A.**, Mehndiratta, M., Chen, Z., Yao, B., & Kayacan, E. (2025). *Adaptive Robust Control Integrated With Gaussian Processes for Quadrotors: Enhanced Accuracy, Fault Tolerance and Anti-Disturbance*. IEEE Transactions on Systems, Man, and Cybernetics: Systems.

4. **Amer, A.**, Mehndiratta, M., Sejersen, J.L.F., Pham, H.X., & Kayacan, E. (2023). *Visual Tracking Nonlinear Model Predictive Control Method for Autonomous Wind Turbine Inspection*. 2023 21st International Conference on Advanced Robotics (ICAR), 431-438.

5. **Amer, A.**, Álvarez-Tuñón, O., Uğurlu, H.İ., Sejersen, J.L.F., Brodskiy, Y., & Kayacan, E. (2023). *UNav-Sim: A visually realistic underwater robotics simulator and synthetic data-generation framework*. 2023 21st International Conference on Advanced Robotics (ICAR), 570-576.

6. **Amer, A.**, Álvarez-Tuñón, O., Falsegar, D., Brodskiy, Y., & Kayacan, E. (2023). *MUDROV: A modular underwater defouling ROV for ship propeller cleaning*. Advanced Marine Robotics Workshop, International Conference on Intelligent Robots and Systems (IROS) 2023.

7. **Amer, A.** (2021). *Shape Optimization of a Flapping Wing for the Atalanta Project FWMAV* (Master's thesis). Delft University of Technology.

Best regards,
Abdelhakim Amer